Lessons

Solutions to Inequalities

When you are working with inequalities, there is always a range of solutions. On a coordinate plane, the solutions are shown by shading a region of the graph.

In a system, more than one inequality is graphed on the same plane. Each inequality has its own region of solutions. The solution to the system is the space where those regions overlap.

Example

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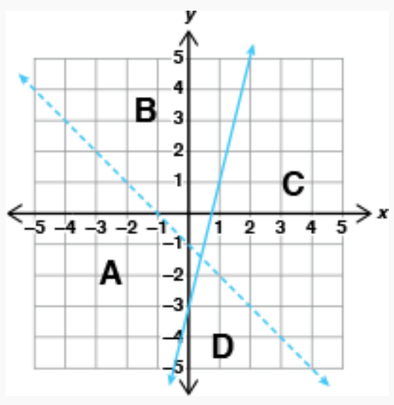
Graph: $\begin{cases} y < -x - 1 \\ y \leq 4x - 3 \end{cases}$

Step 1

Graph both lines on a coordinate plane.

- Use a dashed line if the inequality contains > or <.
- Use a solid line if the inequality contains ≥ or ≤.

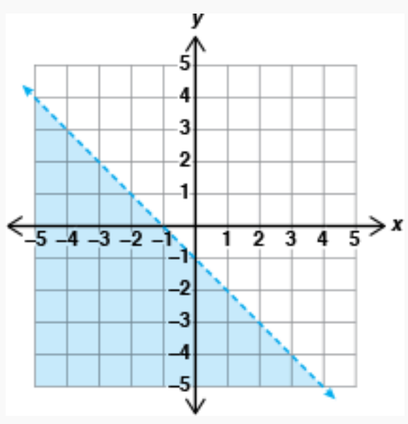
The two lines intersect to create four regions. Here they are labeled A, B, C, and D. One of these regions is the solution to the system.



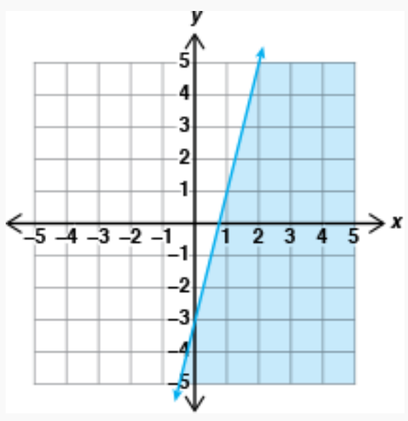
Step 2

Figure out which region is shaded for each inequality. The solution is the space or spaces where the solutions to the individual inequalities overlap.

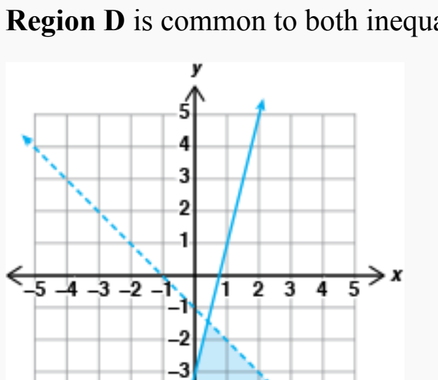
The graph of $y < -x - 1$ includes regions A and D.



The graph of $y \leq 4x - 3$ includes regions C and D.



Region D is common to both inequalities. Any point in this region makes both inequalities true.



Check: You can check your work by substituting a point from the region that you think is the correct one into the original inequalities. Try (1, −3), a point in the shaded region.

y	$<$	$-x - 1$	y	$\leq$	$4x - 3$
-3	$<$	$-(1) - 1$	-3	$\leq$	$4(1) - 3$
-3	$<$	-2 True	-3	$\leq$	1 True

Tip

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Remember, you can find the correct region by testing points. Choose a point from a region and test it in both inequalities. If the point makes both inequalities true, you know this region is part of the solution. Continue until you have tested each region.

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Category

Lesson 5: Inequalities

Related Questions:

Practice questions taken related to this topic

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